S SHRIRAM

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SUMMARY

Third-year computer science undergraduate at Easwari Engineering College, pursuing a parallel BS in Data Science at IIT Madras, with a strong base in data structures, machine learning and full-stack development. Enjoys turning real-world problems into working AI products, from clinical screening to fleet routing and post-quantum security, and aims to grow as a software engineer on teams building AI systems people rely on.

EDUCATION

Easwari Engineering College, Chennai — **2024 – 2028**

B.E. Computer Science and Engineering — CGPA: 8.87 / 10

Indian Institute of Technology Madras (online, parallel degree) — **2024 – 2028**

BS in Data Science and Applications · Diploma in Programming completed; Diploma in Data Science in progress — CGPA: 7.52 / 10

PROJECTS

AuraMed – Women's Health Screening Platform | *React, FastAPI, PyTorch, timm, PostgreSQL* — GitHub · Live · Video

- Built AI screening for breast cancer, cervical cytology and PCOS using ResNet-50 and ViT-B/16 image models with Grad-CAM heatmaps; reached 95% accuracy on PCOS (3,800+ ultrasound images) and 93.13% on cervical cytology.
- Fused model output with clinical scoring (Gail model, Rotterdam criteria) into an explained risk score, and built admin, doctor and patient portals with a sign, lock, addendum and release report workflow.
- Won **2nd place (₹25,000)** at AI Ascend 2026, a 24-hour national hackathon backed by Kyndryl and AWS.

Hash Patch – Offline Post-Quantum Crypto Scanner | *Python, Tree-sitter, CycloneDX, React, Electron* — GitHub · Installer · Video

- Built a scanner that parses Java, Python, Go and C with Tree-sitter and uses call-graph reachability to separate crypto that actually runs from dead or test code.
- Exports a signed CycloneDX 1.6 Crypto Bill of Materials and a migration plan to ML-KEM and ML-DSA prioritised with Mosca's theorem, packaged as an air-gapped Electron desktop app.

FSOC-PAT – AI Beacon Tracking for Optical Comms | *Python, PyTorch, OpenCV, FastAPI, Electron* — GitHub · Video

- Built a simulator for aligning free-space optical terminals (ISRO problem statement) that detects a moving beacon with classical CV, a PyTorch CNN or a hybrid, tracked by Kalman/IMM filters and a PID camera servo.
- Benchmarked all three modes on identical trajectories and seeded disturbances, exporting CSV and PDF reports with measured pass/fail requirement checks; packaged as a Windows desktop app.

OptiFlow – Dynamic Route Optimization | *Python, FastAPI, React, MapLibre GL, OSRM* — GitHub · Video

- Built a real-time fleet routing engine using geographic clustering, deadline-first insertion and 2-opt ordering, enforcing hub deadlines, a 7.5-hour driving limit and fuel capacity on real road networks.
- Simulated traffic, rain and road-block events that trigger explained re-routing decisions, with separate dispatcher and driver views.
- Won the **Transportation & Logistics theme** at CMR Hackfest 3.0, a 36-hour national hackathon.

ACHIEVEMENTS

- **1st Place** – CUMTA Urban Mobility Challenge, Chennai (₹15,000): inferred Metro-to-bus transfers from anonymised tap data
- **2nd Place** – AI Ascend 2026, Saveetha Engineering College (₹25,000)
- **Theme Winner** – CMR Hackfest 3.0, Hyderabad (Transportation & Logistics)
- **1st Prize** – Code Crux (Debugging & DSA), NIRAL'26, CEG, Anna University
- **1st Place** – Code Noobies, Carte Blanche '25, MIT Campus, Anna University
- **2nd Prize** – Project Competition, NeXus 7.0 Industry Institution Summit, Easwari Engineering College
- **2nd Place** – Code Mathrix and Pseudo 2 Code, Mathrix'25, CEG, Anna University
- **3rd Place** – Reverse Coding, Tech Fiesta'25, Chennai Institute of Technology
- 500+ problems solved on LeetCode; CBSE Class X school topper (97.4%), Class XII school rank 3 (95.8%)

CERTIFICATIONS

- **Oracle Certified Professional: Java SE 17 Developer** (1Z0-829), Mar 2026
- **AWS Certified:** Solutions Architect – Associate, Developer – Associate, AI Practitioner, Cloud Practitioner (Jul 2026)
- **NPTEL (IITs):** Management Information Systems and Industry 4.0 & IIoT (Elite + Silver); Blockchain, NLP, Human-Computer Interfaces, Problem Solving in C (Elite); Discipline Star – CSE

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, TypeScript
AI / ML: PyTorch, timm, OpenCV, CNNs, Vision Transformers, Grad-CAM, NLP, RAG
Web & Backend: FastAPI, Flask, React, Vite, Tailwind, NestJS, REST APIs
Data & Cloud: PostgreSQL, SQLite, MySQL, MongoDB, AWS (EC2, API Gateway, Amplify)
Tools: Git, GitHub, Electron, Claude Code (AI-assisted development)